NeuroScope 3D · case BRATS_002

Report 2026-10-17 05:43 UTC
Study ref a6c9e9

Research and teaching prototype. Outputs are suspected tumour regions predicted by the model and estimated volumes. Not a diagnosis; not clinically validated; not for clinical use.

1 This case

Input: one 4-channel file **BRATS_002.nii.gz** (channels FLAIR, T1, T1Gd, T2); grid 240 × 240 × 155 voxels at 1.00 × 1.00 × 1.00 mm, RAS. Expert mask: supplied. Suitability warnings: none.

ESTIMATED VOLUME OF THE SUSPECTED TUMOUR REGION PREDICTED BY THE MODEL

67.74 mL
WT · Whole tumour (codes 1 + 2 + 4)
67,738 voxels × 1.000 mm³ = 67,738 mm³

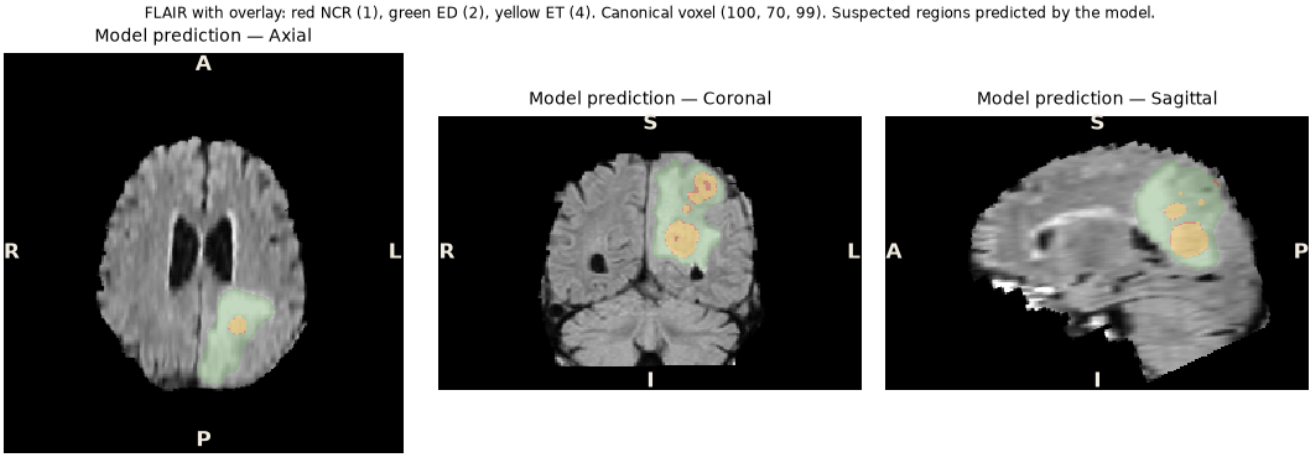
9.70 mL
TC · Tumour core (codes 1 + 4)
9,702 voxels × 1.000 mm³ = 9,702 mm³

7.58 mL
ET · Enhancing tumour (codes 4)
7,584 voxels × 1.000 mm³ = 7,584 mm³

V = N × Δx·Δy·Δz = N × 1.000 × 1.000 × 1.000 mm³ = N × 1.000 mm³; 1 mL = 1,000 mm³. N = voxels the model labelled with the region's BraTS codes.

Region	Codes	Model voxels	Model mL	Expert mL	Dice	Sensitivity
WT Whole tumour	1+2+4	67,738	67.74	63.06	0.920	0.954
TC Tumour core	1+4	9,702	9.70	9.32	0.850	0.867
ET Enhancing tumour	4	7,584	7.58	8.22	0.873	0.839
NCR Necrotic core	1	2,118	2.12	1.11	—	—
ED Oedema	2	58,036	58.04	53.74	—	—

Dice and sensitivity compare the model with the expert mask supplied for this case only (computed for WT, TC and ET).



FLAIR with the model's predicted regions: red NCR (1), green ED (2), yellow ET (4). Radiological view (patient right on image left); slices through the centre of the predicted region.

Stability. Re-running the model on the scan mirrored along each axis (4 passes) gives WT between 67.21 and 69.47 mL; 8.2% of WT voxels change between passes.

2 How the model performs on other cases

Dataset-level, not about this case. 20 labelled cases from Medical Segmentation Decathlon Task01_BrainTumour (labelled training list), labels converted to BraTS codes, split by case ID (one case ID is taken as one patient; the release has no patient identifiers). Cases we trained or tuned on: none (pretrained model used as-is).

Region	Dice mean ± SD	Sensitivity mean ± SD	Cases (n)
WT Whole tumour	0.885 ± 0.108	0.932 ± 0.062	20
TC Tumour core	0.851 ± 0.212	0.912 ± 0.092	20
ET Enhancing tumour	0.764 ± 0.274	0.857 ± 0.189	20 (19 for sensitivity)

Evaluation cases: BRATS_013, BRATS_038, BRATS_048, BRATS_084, BRATS_086, BRATS_145, BRATS_169, BRATS_172, BRATS_175, BRATS_220, BRATS_300, BRATS_305, BRATS_315, BRATS_337, BRATS_346, BRATS_377, BRATS_395, BRATS_397, BRATS_413, BRATS_433. Sensitivity is undefined when the expert marked no voxels of a region; those cases are left out of that mean.

Overlap with training data. We did not train or fine-tune the model, so none of these cases were used by us for training or tuning. However, the pretrained MONAI model was trained on BraTS 2018, and Decathlon Task01 is built from BraTS 2016 and 2017 cases, which BraTS 2018 largely re-uses. Decathlon renamed the cases, so we cannot tell which evaluation cases the model saw during its training; most probably overlap. These metrics are therefore likely optimistic and are not an independent held-out estimate.

3 Provenance

Model	MONAI brats_mri_segmentation (SegResNet) v0.5.4, Apache-2.0 (MONAI Consortium); trained by its authors on BraTS 2018 (285 cases; bundle authors' own split)
Preprocessing	Reorient to RAS, resample to 1 mm, z-score each sequence; SegResNet sliding-window inference; probabilities resampled back to the input grid, threshold 0.5, written as BraTS codes 1 / 2 / 4.
Input SHA-256	`BRATS_002.nii.gz: f4f32cd8d9c46ae672d1e2388b48302bec8b2cbcc334bda3895b0f978d8b5ae2`
Code commit	`09aefc29326b79b7727b412e4f94552ed50c1fa3`
Inference	2026-10-17T05:42:54+00:00 UTC on cuda roi=(240, 240, 160), 9.9 s
Evaluation run	2026-10-17T00:18:40+00:00, commit db28f88841d6
Report	Fixed template in app/services/report.py filled with values recorded by this platform; figure drawn with matplotlib. No generative AI writes or illustrates this report. The segmentation itself is AI model output.

4 Limits

1. Research and teaching prototype. Not a diagnostic tool, not clinically validated; no diagnosis, grading, prognosis or treatment advice.
2. Model trained on BraTS 2018 (adult, pre-operative glioma; skull-stripped, co-registered, 1 mm). Other tumour types, post-operative or non-skull-stripped scans may fail.
3. Volumes are voxel counts of the predicted regions times voxel volume, so they carry every segmentation error. ET and NCR are the least reliable regions.
4. Dataset-level scores are probably optimistic: the evaluation cases likely overlap the model's training data.